

Heaven's Light is Our Guide

Rajshahi University of Engineering & Technology
Department of Computer Science & Engineering



LAB MANUAL

Course No. : **EEE 1252**

Course Name : **Electronic Devices and Circuits Sessional**

Semester : **1st Year Even Semester**

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Laboratory Guidelines

1. Before Coming to Lab

- **Thoroughly Study the Manual:** Read and understand the theory, objectives, and procedure of the experiment to be performed *before* the lab session.
- **Pre-Lab Preparation:** Answer the "Post-Lab Questions" from the previous experiment and be prepared to discuss them. For the current experiment, try to derive the theoretical formulas (e.g., gain, frequency) beforehand.
- **Understand the Circuit:** Analyze the circuit diagram provided. Identify all components, their values, and their connections.
- **Plan Your Work:** Mentally walk through the experimental steps. Note down the specific measurements you need to take (e.g., voltages, currents, waveforms).
- **Bring Necessary Items:** Carry the lab manual, a notebook for rough work and data, a pen, a calculator, and any required pre-lab assignments.

2. During Lab Session

A. Conduct

- **Punctuality and Attendance:** Be on time. Your presence and active participation are crucial.
- **Follow Instructions:** Listen carefully to the instructor's or teaching assistant's (TA's) briefing and follow all their instructions.
- **Maintain Discipline:** Keep the lab bench tidy. Avoid unnecessary conversation and noise that can distract others.
- **Handle with Care:** Treat all instruments (oscilloscopes, multimeters, power supplies), components, and project boards with care. They are delicate and expensive.

B. Work Process

- **Check Components First:** Before connecting, use a multimeter to verify the values of resistors, check diodes and transistors for functionality, and ensure capacitors are not shorted.
- **Neat and Correct Wiring:** Build the circuit methodically on the project board. Use wires of appropriate lengths and colors (e.g., red for Vcc, black for ground) for clarity. Double-check all connections against the circuit diagram before applying power.
- **Power Off During Wiring:** Always ensure the DC power supply is *OFF* and its voltage is set to zero while making or modifying connections.
- **Use Proper Settings:** Set instruments correctly. For example, use the DC coupling mode on the oscilloscope for DC measurements and AC coupling for observing AC waveforms.
- **Apply Power Gradually:** After verification, turn on the power supply and gradually increase the voltage to the required level as per the procedure.
- **Methodical Data Collection:** Take readings systematically and record them *immediately* in your data tables. Note the units for all measurements.

3. After Completing Experiment

- **Power Down Safely:** First, turn the DC power supply voltage to zero, then switch it off before disassembling the circuit.
- **Disassemble Neatly:** Carefully disconnect all wires and components. Return them to their proper places.
- **Verify Results:** Do a quick check to see if your recorded data and observed waveforms make sense according to the theory.
- **Get Your Data Signed:** Have the instructor check your raw data and initial your lab notebook/manual before you leave the lab station.

4. Lab Report Writing

- **Timely Submission:** Submit a well-structured and complete lab report by the specified deadline.
- **Standard Format:** Follow the structure outlined in the manual for each experiment:

- **Title, Objectives, and Theory** (in your own words).
- **Circuit Diagram** (neatly drawn or pasted).
- **Experimental Data/Results:** Present all recorded data in clearly labeled tables. Attach printed or neatly drawn input/output waveforms, labeling key parameters (e.g., V_{p-p} , time period).
- **Calculations:** Show sample calculations for all required parameters (e.g., static/dynamic resistance, voltage gain, frequency, % error).
- **Discussion and Conclusion:** This is a critical section.
 - **Discussion:** Interpret your results. Explain the shape of the waveforms. Comment on the agreement between theoretical and experimental values. Analyze the sources of errors (e.g., component tolerances, instrument inaccuracies, parasitic effects).
 - **Conclusion:** Summarize the key findings of the experiment and state whether the objectives were met.
- **Post-Lab Questions:** Answer all questions thoughtfully and completely.

5. General Rules & Precautions

- **Safety First:** Be aware of the power supply limits. Avoid short-circuiting outputs. Report any damaged equipment or exposed wires immediately.
- **Correct Identification:** Always properly identify the terminals of components (BJT - Base, Collector, Emitter; FET - Gate, Drain, Source; Op-Amp; Diode Anode/Cathode) before connecting.
- **Calibrate Instruments:** Ensure oscilloscopes and multimeters are properly calibrated before taking measurements.
- **Capacitor Discharge:** Always discharge large capacitors (like the 10 μ F in rectifier circuits) safely with a resistor before handling the circuit after power-off.
- **No Food or Drink:** Do not bring food or drinks into the laboratory.

6. Evaluation

Students will be graded on:

Lab report	25 marks
Lab final	20 marks
Quiz	20 marks
Attendance	10 marks

7. Lab Curriculum

Sessional on Electronic Drives and Circuits: Diode, Bipolar Junction Transistor (BJT), Field Effect Transistor (FET), Op-Amps, 555 Timer, Filter Circuit, Logic Gate Design using BJT.

Table-1: CO-PO mapping for EEE 1252

CO No.	CO Statement	Mapped PO	Domain/ level of Learning Taxonomy	Delivery Methods and Activities	Assessment Tools
CO1	Development of different types of amplifiers, operational, and multivibrator circuits using semiconductor devices and analog ICs.	PO3	Psychomotor (Perfect)	Lecture Discussion Interaction Lab Manual Experiment	Quiz Lab Report
CO2	Investigate the characteristics of different types of semiconductor devices.	PO4	Psychomotor (Perfect)	Lecture Discussion Interaction Lab Manual Experiment	Quiz Lab Report
CO3	Using modern engineering tools and techniques to design and implement electronic circuits and systems.	PO5	Psychomotor (Articulate)	Lecture Discussion Interaction Lab Manual Experiment	Viva Lab Final

Table-2: PO statements

PO1	Engineering knowledge: Apply knowledge of mathematics, natural science, engineering fundamentals and an engineering specialization respectively to the solution of complex engineering problems.
PO2	Problem analysis: Identify, formulate, research literature and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences.
PO3	Design/development of solutions: Design solutions for complex engineering problems and design systems, components or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.
PO4	Investigation: Conduct investigations of complex problems using research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.
PO5	Modern tool usage: Create, select and apply appropriate techniques, resources, and modern engineering and IT tools, including prediction and modelling, to complex engineering problems, with an understanding of the limitations.
PO6	The engineer and society: Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice and solutions to complex engineering problems.
PO7	Environment and sustainability: Understand and evaluate the sustainability and impact of professional engineering work in the solution of complex engineering problems in societal and environmental contexts.
PO8	Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of engineering practice.
PO9	Individual work and teamwork: Function effectively as an individual, and as a member or leader in diverse teams and in multi-disciplinary settings.
PO10	Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO11	Project management and finance: Demonstrate knowledge and understanding of engineering management principles and economic decision-making and apply these to one’s own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO12	Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

List of the Experiments

Sl. No.	Name of Experiment	Corr. CO	Page
1	Study of I-V characteristics of a p-n junction diode	CO2	02 - 04
2	Study of Half-wave and Full-wave rectifier circuit using diode	CO1	05 - 07
3	Study of Zener Diode as a voltage regulator	CO2	08 - 09
4	Study of BJT biasing circuit	CO2	10 -12
5	Study of single-stage BJT amplifier circuit	CO2	13 - 14
6	Study of single-stage FET amplifier circuit	CO2	15 - 16
7	Study of transistor switching network using BJT	CO2	17 – 18
8	Study of Inverting and Non-Inverting amplifier using Op-Amp	CO1	19 - 21
9	Study of differentiator and integrator circuit using Op-Amp	CO1	22 -2 4
10	Study of the astable multivibrator operation using 555 timer	CO1	25 – 26
11	Open-Ended Lab: Study of constructing digital logic circuit using semiconductor devices	CO1	27 - 27

Experiment No.: 01

Name of the experiment: Study of I-V characteristics of a p-n junction diode.

1.1. Objectives

- To observe and draw the forward and reverse bias I-V Characteristics of a p-n Junction diode.
- To calculate static and dynamic resistance in both forward and reverse bias condition.

1.2. Theory

A p-n junction diode is a fundamental semiconductor device formed by joining p-type and n-type materials. The I-V characteristics of a diode describe the relationship between the applied voltage and the resulting current, providing insights into its behavior in different biasing conditions.

Biasing of p-n Junction:

- Forward Bias:** When the p-side is connected to the positive terminal of a power source and the n-side to the negative terminal, the external voltage reduces the built-in potential barrier. This allows charge carriers to move across the junction, leading to a significant increase in current beyond a certain threshold voltage ($\sim 0.7\text{V}$ for silicon and $\sim 0.3\text{V}$ for germanium diodes).
- Reverse Bias:** When the p-side is connected to the negative terminal and the n-side to the positive terminal, the potential barrier increases, preventing majority carrier flow. Only a small leakage current, due to minority carriers, flows. If the reverse voltage exceeds a critical breakdown value, the diode undergoes **breakdown** (Zener or avalanche), leading to a sharp increase in current.

I-V Characteristics:

The current-voltage relationship of a p-n junction diode is given by the Shockley equation:

$$I_D = I_s \left(e^{\frac{V}{nV_T}} - 1 \right)$$

where:

- I_D is the diode current,
- I_s is the reverse saturation current,
- V is the applied voltage,
- n is the ideality factor (typically between 1 and 2),
- V_T is the thermal voltage (approximately 25.9mV at room temperature).

In **forward bias**, the current increases exponentially with voltage after overcoming the threshold voltage. In **reverse bias**, the current remains minimal until breakdown occurs.

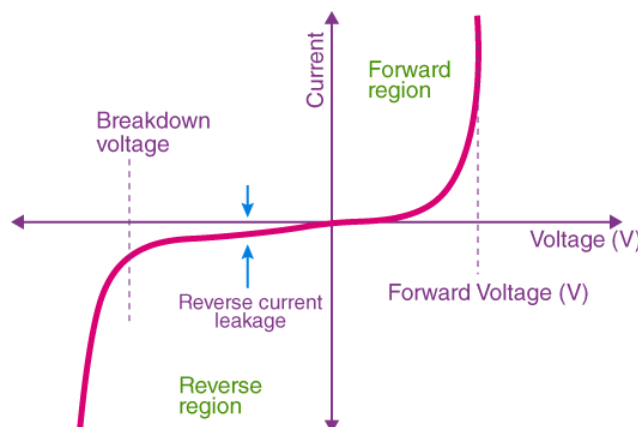


Fig. 1.1. I-V characteristics of a p-n junction diode.

1.3. Required Apparatus

- Diode (1N4007, 1 piece)
- Resistor (1kΩ: 1 piece)
- DC supply (0 – 15 V)
- Ammeter (0 – 2A)
- Voltmeter (0 – 5V)
- Project board
- Connecting wires

1.4. Circuit Diagram

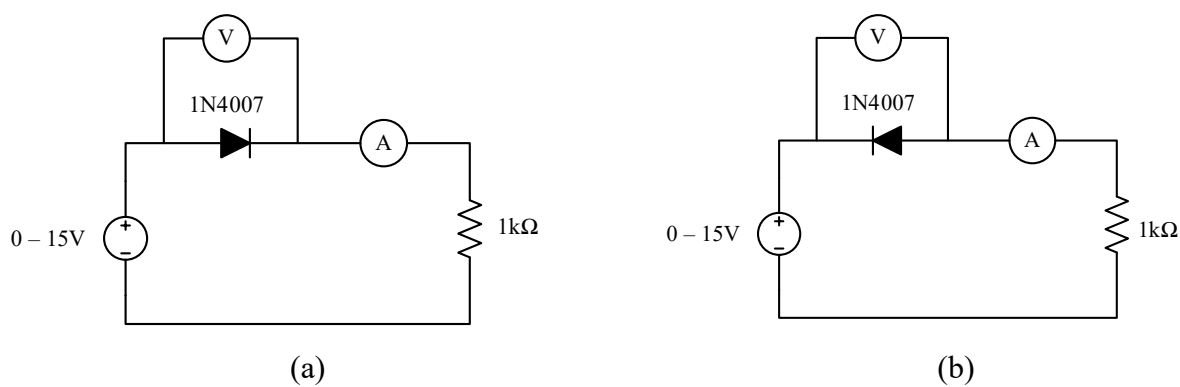


Fig. 1.2. Circuit diagram for determining I-V characteristics at (a) forward bias and (b) reverse bias condition of a p-n junction diode.

1.5. Experimental Procedure

1. Connect the circuit according to Fig.1.2 (a) such that the p-terminal connects to the positive and n-terminal to the negative of the power supply.
2. Gradually increase the voltage in small steps and measure the corresponding current using the ammeter.
3. Record the readings until the diode conducts significantly (~0.7V for silicon).
4. Next, connect the circuit according to Fig.1.2 (b) so that the p-terminal is connected to the negative and n-terminal to the positive of the power supply.
5. Increase the reverse voltage in small steps and measure the reverse current using the ammeter.
6. Record the values until breakdown voltage (if applicable) is approached.
7. Plot Voltage (V) vs. Current (I) on a graph for both forward and reverse bias.

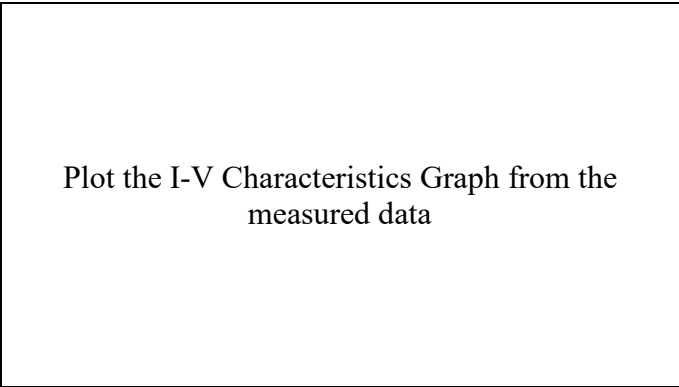
1.6. Experimental Results

Table-1: Measured data for forward bias mode.

SL. No.	Applied Voltage (V)	Forward Voltage (V)	Forward Current (mA)

Table-2: Measured data for reverse bias mode.

SL. No.	Applied Voltage (V)	Reverse Voltage (V)	Reverse Current (mA)
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1.7. Discussion and Conclusion

State the concluding remarks on the experiment.

1.8. Precautions

- All the connections should be correct.
- Parallax error should be avoided while taking the readings from the Analog meters.

1.9. Post-Lab Questions

1. What is depletion region of a diode?
2. What is meant by transition & space charge capacitance of a diode?
3. Is the V-I relationship of a diode Linear or Exponential?
4. Define cut-in voltage of a diode and specify the values for Si and Ge diodes?
5. What are the applications of a p-n diode?
6. Draw the ideal characteristics of P-N junction diode?
7. What is the diode equation?
8. What is PIV?
9. What is the break down voltage?
10. What is the effect of temperature on PN junction diodes?

Experiment No.: 02

Name of the experiment: Study of Half-wave and Full-wave rectifier circuit using diode

2.1. Objectives

- To implement the half-wave and full-wave rectifier using a p-n Junction diode.
- To observe the output waveforms of the half-wave and full-wave rectifier.

2.2. Theory

A rectifier converts an AC signal into a DC signal. From the characteristic curve of a diode, we observe that it allows the current to flow when it is in the forward bias only. In the reverse bias it remains open. So, when an alternating voltage is applied across a diode it allows only the half cycle (positive half depending on orientation of diode in the circuit) during its forward bias condition, other half cycle will be clipped off. In the output the load will get DC signal.

Diode rectifier can be categorized in two major types. They are:

- i. Half-wave rectifier and
- ii. Full-wave rectifier

Half-wave rectifier: Half-wave rectifier circuit can be built by using a single diode. In this circuit the load receives approximately half of the input power, Average dc voltage is low, Due to the presence of ripple output voltage is not smooth one. The circuit diagram and the wave shape of the input and output voltage of a half-wave rectifier are shown below.

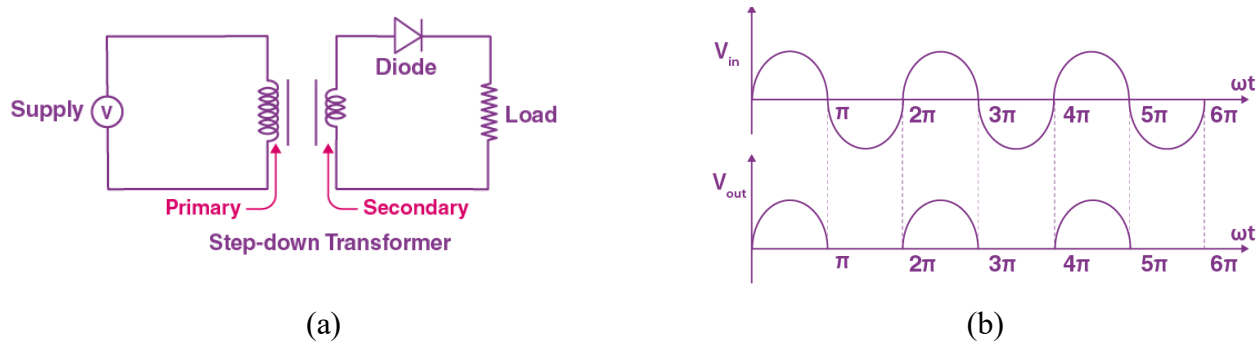


Fig. 2.1. Half-wave rectifier (a) circuit and (b) corresponding waveforms.

Full-wave rectifier: In the full-wave rectifier both of the half cycle is present in the output. Two circuits are used as full-wave rectifiers. They are:

- a) Full-wave rectifier using center-tapped transformer and
- b) Full-wave bridge rectifier.

Using a center-tapped transformer: Two diodes will be connected to the ends of the transformer and the load will be between the diode and the center tap.

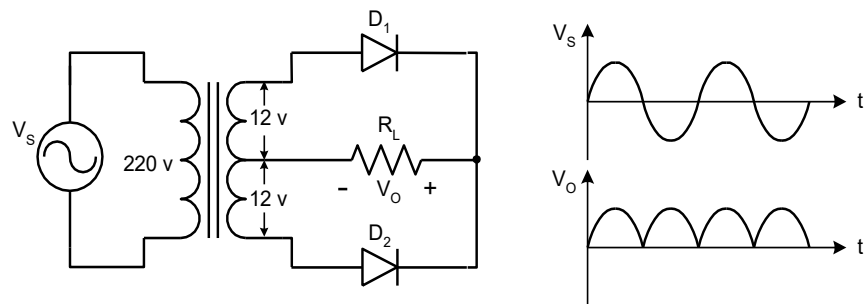


Fig. 2.2. Full-wave rectifier using center tapped transformer.

This circuit has some advantages over the half-wave rectifier:

- Wastage of power is less,
- Average DC output increase significantly,
- Wave shape becomes smoother.

But the disadvantage is, it requires more space and becomes bulky because of the transformer.

Bridge rectifier: A bridge rectifier overcomes all disadvantages described above.

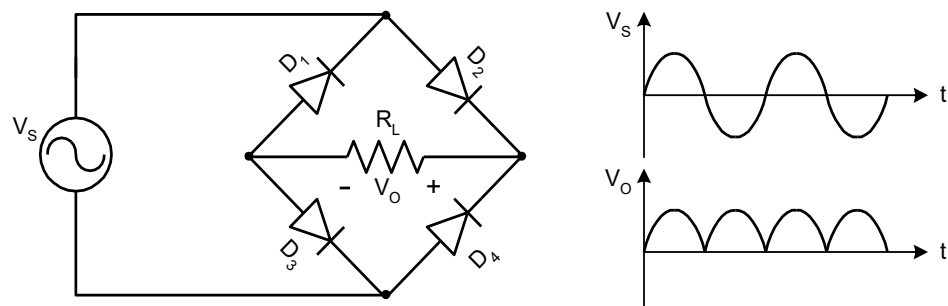


Fig. 2.3. Full-wave bridge rectifier.

2.3. Required Apparatus

- Diode (1N4007, 4 pieces)
- Resistor (1k Ω : 1 piece)
- Capacitor (1 μ F, 10 μ F: 1 piece each)
- Center tapped transformer (1 piece)
- DC supply (0 – 15 V)
- Signal Generator
- Oscilloscope
- Project board
- Multimeter
- Connecting wires

2.4. Circuit Diagram

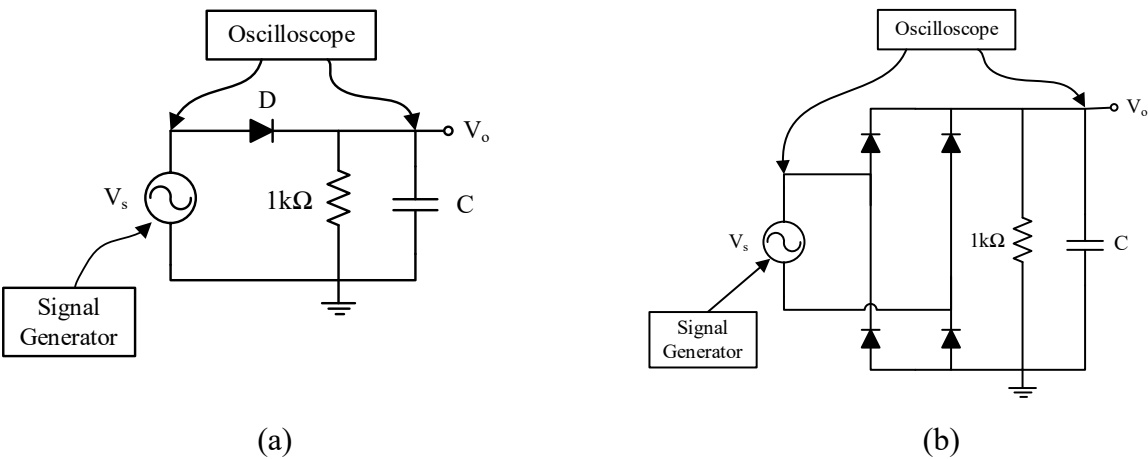


Fig. 2.4. Circuit diagram for (a) half-wave and (b) full-wave bridge rectifier circuit.

2.5. Experimental Procedure

1. Connect the circuit according to Fig. 2.4 (a) without capacitor.

2. Observe and record the input and output voltage waveforms in the oscilloscope.
3. Connect a $1\mu\text{F}$ capacitor and repeat step 2. Observe how the output waveform changes.
4. Connect the $10\mu\text{F}$ capacitor and repeat step 2. Observe how the output waveform changes.
5. Next, connect the circuit according to Fig. 2.4 (b) without capacitor.
6. Observe and record the input and output voltage waveforms in the oscilloscope.
7. Connect a $1\mu\text{F}$ capacitor and repeat step 6. Observe how the output waveform changes.
8. Connect the $10\mu\text{F}$ capacitor and repeat step 6. Observe how the output waveform changes.

2.6. Experimental Results

Attach the recorded input and output waveforms

Briefly explain the observations from the various output waveforms.

2.7. Discussion and Conclusion

State the concluding remarks on the experiment.

2.8. Precautions

- All the connections should be correct.
- Oscilloscope must be calibrated properly.

2.9. Post-Lab Questions

1. What is meant by rectifier?
2. Which type of transformer is used for the rectifier input?
3. How ripples in the output are mitigated in a rectifier circuit?
4. Why capacitor is used across load in a rectifier circuit?

Experiment No.: 03

Name of the experiment: Study of Zener Diode as a voltage regulator

3.1. Objectives

- To implement Zener diode as a voltage regulator
- To observe the output waveforms of the voltage regulator circuit

3.2. Theory

A Zener diode is a special type of semiconductor diode designed to allow current to flow in the reverse direction when the voltage exceeds a specific breakdown voltage, known as the *Zener voltage* (V_Z). Unlike regular diodes, which block reverse current beyond a certain limit, Zener diodes are designed to operate in breakdown mode without damage, making them suitable for *voltage regulation* applications.

Zener Diode as a Voltage Regulator: A voltage regulator maintains a stable output voltage despite variations in input voltage or load current. The Zener diode can be used as a shunt voltage regulator, where it is connected in reverse bias across the load.

- **Normal Operation (No Breakdown):**
 - When the input voltage (V_{in}) is less than the Zener voltage (V_Z), the Zener diode remains in its off state and acts as an open circuit.
 - The entire input voltage appears across the load.
- **Zener Breakdown Region:**
 - When V_{in} exceeds V_Z , the diode enters breakdown mode and maintains a nearly constant voltage (V_Z) across the load.
 - The excess input voltage is dropped across a series resistor, which limits the current through the diode and prevents damage.

3.3. Required Apparatus

- Zener diode (1 piece)
- Resistor (1 k Ω , 10 k Ω : 1 piece each)
- DC Power supply (0 – 15V)
- Project board
- Multimeter
- Connecting wires

3.4. Circuit Diagram

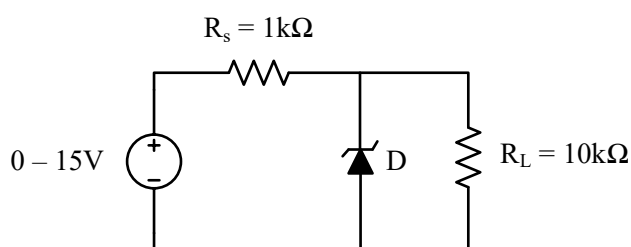


Fig. 3.1. Circuit diagram for implementing Zener diode as voltage regulator.

3.5. Experimental Procedure

1. Connect the circuit according to Fig. 3.1.

2. Increase the input voltage gradually and record the corresponding voltage across the load.
3. Repeat step-2 until the voltage across the zener diode exceeds its zener voltage.
4. Observe if the load voltage remains constant when the voltage across the zener diode exceeds its zener voltage.

3.6. Experimental Results

Table-1: Data table for observing the operation of zener diode as a voltage regulator.

SL. No.	Input Voltage V_{in} (V)	Voltage across series Resistor, V_{Rs} (V)	Voltage across load V_L (V)

3.7. Discussion and Conclusion

State the concluding remarks on the experiment.

3.8. Precautions

- All the connections should be correct.
- Parallax error should be avoided while taking the readings from the Analog meters.

3.9. Post-Lab Questions

1. What is the purpose of the series resistor (R_s) in the voltage regulator circuit?
2. How does the Zener diode maintain a constant output voltage despite variations in input voltage?
3. If the Zener voltage is 5.6V and the input voltage is 12V, how would you calculate the required series resistor value for a maximum Zener current of 20mA?
4. What are the practical limitations of using a Zener diode as a voltage regulator in high-power applications?

Experiment No.: 04

Name of the experiment: Study of BJT biasing circuit

4.1. Objectives

- To implement BJT biasing circuits
- To set the operating point of a BJT at the mid-point of the characteristics curve

4.2. Theory

A Bipolar Junction Transistor (BJT) is a three-terminal semiconductor device used for amplification and switching applications. To ensure its proper operation, the transistor must be biased in the desired region of operation—either cutoff, active, or saturation. The active region is used for amplification, while the saturation and cutoff regions are used in switching applications. There are different types of biasing circuits in which different parameters can be used to change the operating point.

Types of BJT Biasing Circuits

1. Fixed Bias Circuit

- A simple biasing method where a resistor is connected between the base and the DC supply voltage.
- Easy to implement but has poor thermal stability.

2. Collector-to-Base Bias

- A resistor is connected between the collector and the base to improve stability.
- Provides better thermal stability than fixed bias but still has limitations.

3. Voltage-Divider Bias

- A resistive voltage divider network is used to provide stable base biasing.
- Offers better Q-point stability and is widely used in amplifier circuits.

Among the above-mentioned biasing circuits, voltage-divider circuit provides better stability of the operating point. Hence, it is preferred over other biasing techniques.

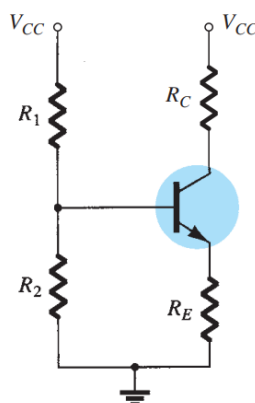


Fig. 4.1. BJT voltage-divider biasing circuit.

From the circuit shown in Fig.4.1, following equations are obtained.

$$R_{Th} = R_1 \parallel R_2 \quad (1)$$

$$E_{Th} = V_{R_2} = \frac{R_2 V_{CC}}{R_1 + R_2} \quad (2)$$

$$I_B = \frac{E_{Th} - V_{BE}}{R_{Th} + (\beta + 1)R_E}$$

(3)

$$V_{CE} = V_{CC} - I_C(R_C + R_E)$$

(4)

4.3. Required Apparatus

- BJT (BD135, 1 piece)
- Resistors (1 kΩ, 2.2 kΩ, 4.7 kΩ: 1 piece each)
- Potentiometer (100 kΩ: 1 piece)
- DC Power supply (0 – 15V)
- Project board
- Multimeter
- Connecting wires

4.4. Circuit Diagram

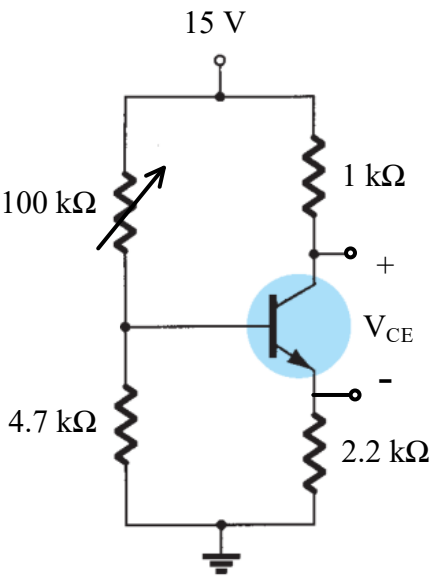


Fig. 4.1. Circuit diagram for BJT voltage divider biasing circuit.

4.5. Experimental Procedure

1. Connect the circuit according to Fig. 4.1.
2. Gradually vary the potentiometer so that V_{CE} becomes equal to the $V_{CC}/2$
3. Measure V_{RC} , V_{BE} , I_B , I_C and β
4. Compare the measured values with the theoretical values.

4.6. Experimental Results

Table-1: Data table for comparing the measured results with the theoretical results.

	Collector-Emitter Voltage, V_{CE} (V)	Collector Resistor Voltage, V_{RC} (V)	Base-Emitter Voltage, V_{BE} (V)	Base Current, I_B (μA)	Collector Current, I_C (mA)
Experimental Data					
Theoretical Data					
%error					

4.7. Discussion and Conclusion

State the concluding remarks on the experiment.

4.8. Precautions

- All the connections should be correct.
- Transistor terminals must be identified properly
- Calibration must be performed properly

4.9. Post-Lab Questions

1. What is the importance of biasing in a BJT circuit?
2. How does temperature variation affect the Q-point of a transistor?
3. Why is voltage-divider bias preferred over other biasing techniques?
4. How does the stability factor affect transistor biasing?
5. Based on your experimental results, how closely did your measured Q-point match the theoretical predictions? What factors might cause discrepancies?

Experiment No.: 05

Name of the experiment: Study of single-stage BJT amplifier circuit

5.1. Objectives

- To implement a single-stage BJT amplifier circuit
- To observe the input and output waveforms of a BJT amplifier circuit

5.2. Theory

A single-stage BJT amplifier is a fundamental building block in analog electronics. It amplifies weak AC signals while maintaining fidelity. The most commonly used configuration is the common-emitter amplifier, which provides voltage gain and is widely used in signal amplification applications. The BJT operates in the active region where it functions as an amplifier. A biasing network is used to establish a proper Q-point (operating point) for stable operation. Voltage divider biasing network is mostly preferred as it provides better stability of operating point. An AC coupling capacitor is used to block DC components from the input signal.

5.3. Required Apparatus

- BJT (BD135, 1 piece)
- Resistors (1 k Ω , 2.2 k Ω , 4.7 k Ω : 1 piece each)
- Potentiometer (100 k Ω , 50 k Ω : 1 piece each)
- DC Power supply (0 – 15V)
- Project board
- Multimeter
- Connecting wires

5.4. Circuit Diagram

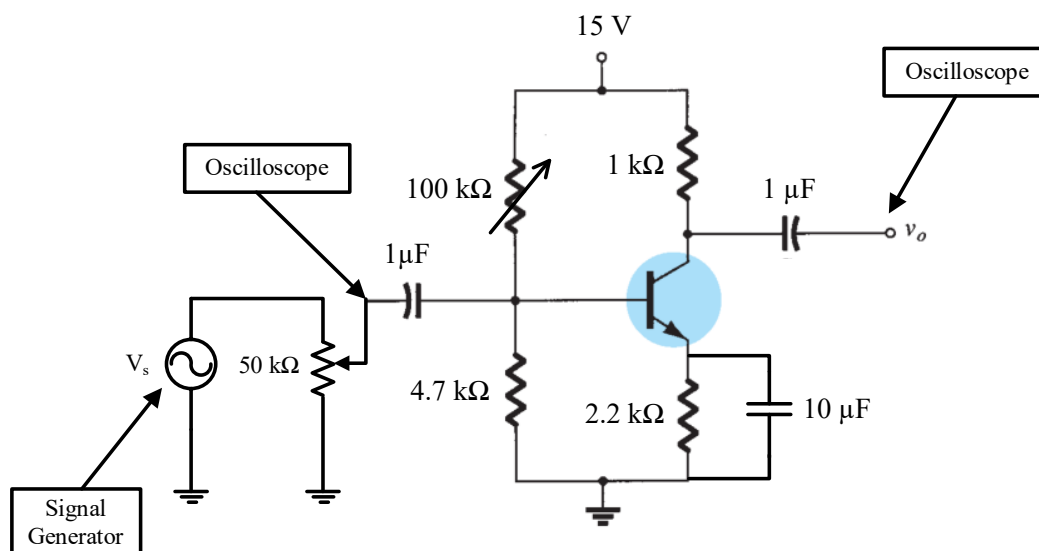


Fig. 5.1. Circuit diagram for implementing single-stage BJT amplifier.

5.5. Experimental Procedure

1. Connect the circuit according to Fig. 5.1 without the capacitors.
2. Perform the mid-point biasing i.e., $V_{CE} = V_{CC}/2$ by tuning the potentiometer.
3. After the completion of biasing, connect the coupling capacitors and bypass capacitors.
4. Supply a small signal to the input side using signal generator and record the corresponding output waveform in the oscilloscope.

5. Compare the experimental and theoretical voltage gain of the amplifier circuit.

5.6. Experimental Results

Attach the input and output waveforms

Calculation:

Theoretical Gain,

$$A_v = -\frac{R_C}{R_E}$$

Measured Gain,

$$A'_v = \frac{V_{out} (peak-to-peak)}{V_{in} (peak-to-peak)}$$

$$\% \text{ Error} = \frac{|A_v - A'_v|}{A_v} \times 100\%$$

5.7. Discussion and Conclusion

State the concluding remarks on the experiment.

5.8. Precautions

- All the connections should be correct.
- Transistor terminals must be identified properly
- Calibration must be performed properly

5.9. Post-Lab Questions

1. What is the purpose of the biasing circuit in a BJT amplifier?
2. Why does the amplifier introduce a phase shift of 180° ?
3. How does adding an emitter bypass capacitor affect the amplifier's gain and bandwidth?
4. What factors influence the input and output impedance of a BJT amplifier?

Experiment No.: 06

Name of the experiment: Study of single-stage FET amplifier circuit

6.1. Objectives

- To implement a FET amplifier circuit
- To observe the input and output waveforms of a FET amplifier circuit
- To compare the experimental results with the theoretical results

6.2. Theory

A Field Effect Transistor (FET) is a voltage-controlled semiconductor device that operates with high input impedance and low output impedance, making it suitable for amplification applications. A single-stage FET amplifier consists of a single transistor stage that provides amplification of the input signal. The FET has some advantages and some disadvantages relative to the bipolar transistor. Field-effect transistors are preferred for weak-signal work, for example in wireless, communications and broadcast receivers. They are also preferred in circuits and systems requiring high impedance. The FET is not, in general, used for high-power amplification, such as is required in large wireless communications and broadcast transmitters. A common source JFET amplifier has high input impedance and a moderate voltage gain. Also, the input and output voltages are 180 degrees out of Phase.

6.3. Required Apparatus

- JFET (BFW10, 1 piece)
- Resistors (1 M Ω : 1 piece, 1 k Ω : 2 pieces)
- Potentiometer (100 k Ω : 1 piece)
- Capacitors (10 μ F: 1 piece, 100 μ F: 2 pieces)
- DC Power supply (0 – 15V)
- Signal Generator
- Oscilloscope
- Project board
- Multimeter
- Connecting wires

6.4. Circuit Diagram

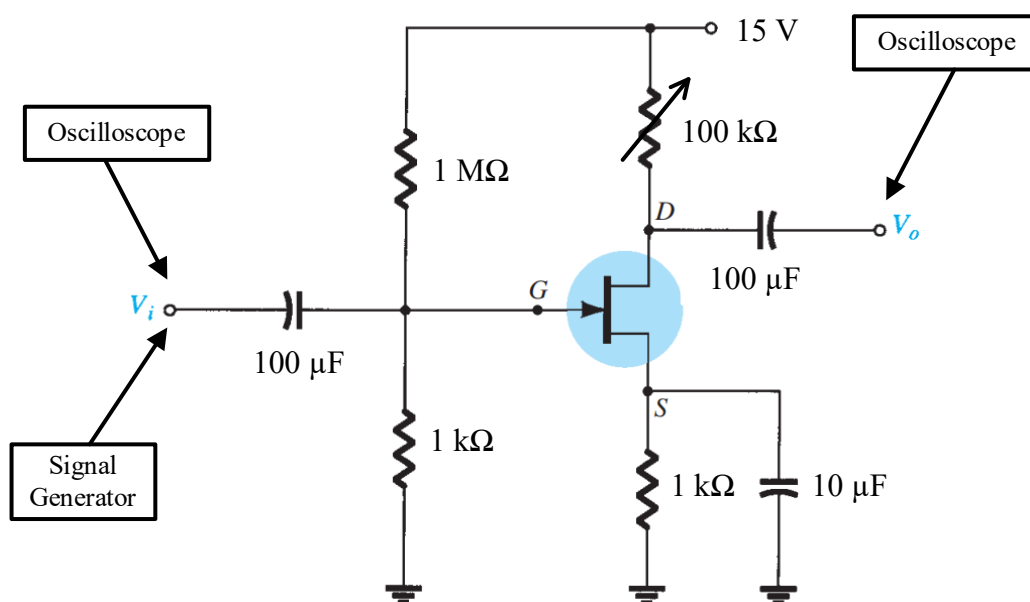
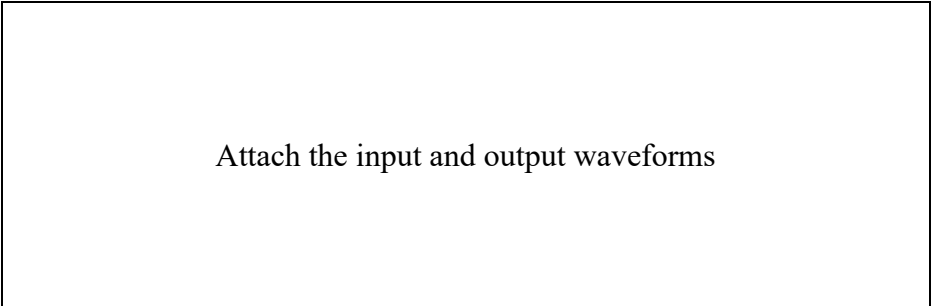


Fig. 6.1. Circuit diagram for implementing single-stage common source JFET amplifier.

6.5. Experimental Procedure

- 1. Connect the circuit according to Fig. 5.1 without the capacitors.
- 2. Perform the mid-point biasing.
- 3. After the completion of biasing, connect the coupling capacitors and bypass capacitors.
- 4. Supply a small signal to the input side using signal generator and record the corresponding output waveform in the oscilloscope.
- 5. Compare the experimental and theoretical voltage gain of the amplifier circuit.

6.6. Experimental Results



Calculation:

Theoretical Gain,

$$A_v = -g_m (R_D || R_L)$$

Measured Gain,

$$A'_v = \frac{V_{out (peak-to-peak)}}{V_{in (peak-to-peak)}}$$

$$\% \text{ Error} = \frac{|A_v - A'_v|}{A_v} \times 100\%$$

6.7. Discussion and Conclusion

State the concluding remarks on the experiment.

6.8. Precautions

- All the connections should be correct.
- Transistor terminals must be identified properly
- Calibration must be performed properly

6.9. Post-Lab Questions

- 1. What is the role of the drain resistor in a single-stage FET amplifier?
- 2. How does the choice of biasing method affect the amplifier’s performance?
- 3. Why does a FET amplifier typically have high input impedance?
- 4. How would increasing the source resistance affect the voltage gain?

Experiment No.: 07

Name of the experiment: Study of transistor switching network using BJT

7.1. Objectives

- To implement the switching circuit using BJT
- To observe the input and output waveforms of a BJT switching circuit

7.2. Theory

For switching applications, the BJT is operated between cutoff and saturation states to act as a digital switch. A BJT switch is controlled by the base current (I_B):

- When $I_B = 0 \rightarrow$ Transistor is in cutoff region $\rightarrow$ No conduction $\rightarrow$ Acts as an open switch.
- When I_B is sufficiently high $\rightarrow$ Transistor enters saturation region $\rightarrow$ Conduction occurs $\rightarrow$ Acts as a closed switch.

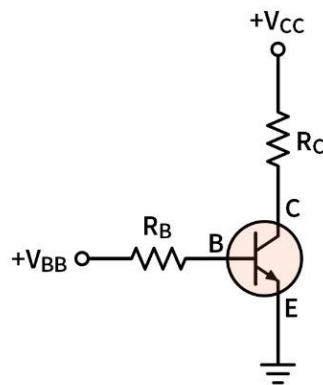


Fig. 7.1. Switching network for BJT.

When a small input voltage (V_{BB}) is applied at the base, the transistor turns **ON** (saturation). The collector-emitter voltage (V_{CE}) drops close to **0V**, allowing current to flow through R_C . When V_{BB} is removed, the transistor turns **OFF** (cutoff), and no current flows through R_C .

7.3. Required Apparatus

- BJT (BD135, 1 piece)
- Resistors (1 k Ω : 2 pieces)
- DC Power supply (0 – 15V)
- Project board
- Multimeter
- Connecting wires

7.4. Circuit Diagram

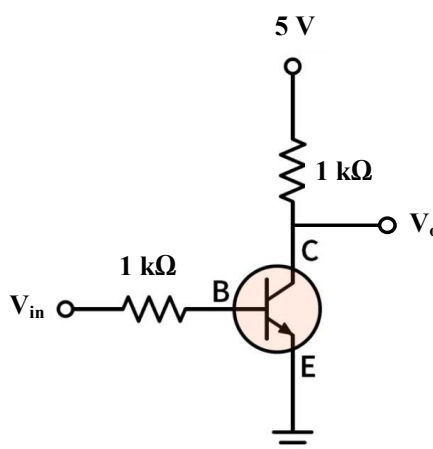


Fig. 7.2. Circuit diagram for implementing BJT as a switch.

7.5. Experimental Procedure

1. Connect the circuit according to Fig. 7.2.
2. Provide $V_{BB} = 5\text{ V}$ and observe the output across V_{CE} (it must be zero).
3. Provide $V_{BB} = 0\text{ V}$ and observe the output across V_{CE} (it must be 5 V).
4. Or, instead of executing step-2 and 3, provide a square wave (having 1 and zero states) using a signal generator to the V_{BB} and observe the output at the V_{CE} terminals at an oscilloscope.

7.6. Experimental Results

Attach the input and output waveforms

7.7. Discussion and Conclusion

State the concluding remarks on the experiment.

7.8. Precautions

- All the connections should be correct.
- Transistor terminals must be identified properly.
- Calibration must be performed properly.

7.9. Post-Lab Questions

1. What are the different operating regions of a BJT? Which two regions are used for switching?
2. Explain the significance of the saturation voltage in a transistor switch.
3. How does the value of the base resistor (R_B) affect the switching speed of the transistor?

Experiment No.: 08

Name of the experiment: Study of Inverting and Non-Inverting amplifier using Op-Amp

8.1. Objectives

- To implement the inverting and non-inverting amplifier using Op-amp
- To observe the input and output waveforms of the inverting and non-inverting amplifier

8.2. Theory

Inverting Amplifier

An inverting amplifier is an Op-Amp configuration where the input signal is applied to the inverting terminal (-), while the non-inverting terminal (+) is grounded. A feedback resistor provides negative feedback, ensuring stability and controlled gain.

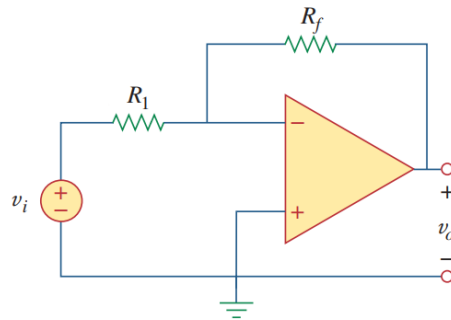


Fig. 8.1. Inverting amplifier circuit using op-amp.

The gain of the inverting amplifier is provided by:

$$v_o = -\frac{R_f}{R_1}v_i$$

Non-Inverting Amplifier

A non-inverting amplifier applies the input signal to the non-inverting terminal (+), while the inverting terminal (-) is connected to a feedback network. This configuration provides a positive gain without phase inversion.

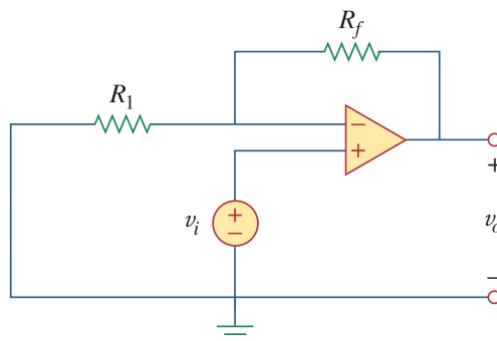


Fig. 8.2. Non-inverting amplifier circuit using op-amp.

The gain of the non-inverting amplifier is provided by:

$$v_o = \left(1 + \frac{R_f}{R_1}\right)v_i$$

8.3. Required Apparatus

- Op-amp (μc741 , 1 piece)
- Resistors (1 k Ω , 10 k Ω : 1 piece each)
- Signal Generator
- Oscilloscope
- DC Power supply (0 – 15V)
- Project board
- Multimeter
- Connecting wires

8.4. Circuit Diagram

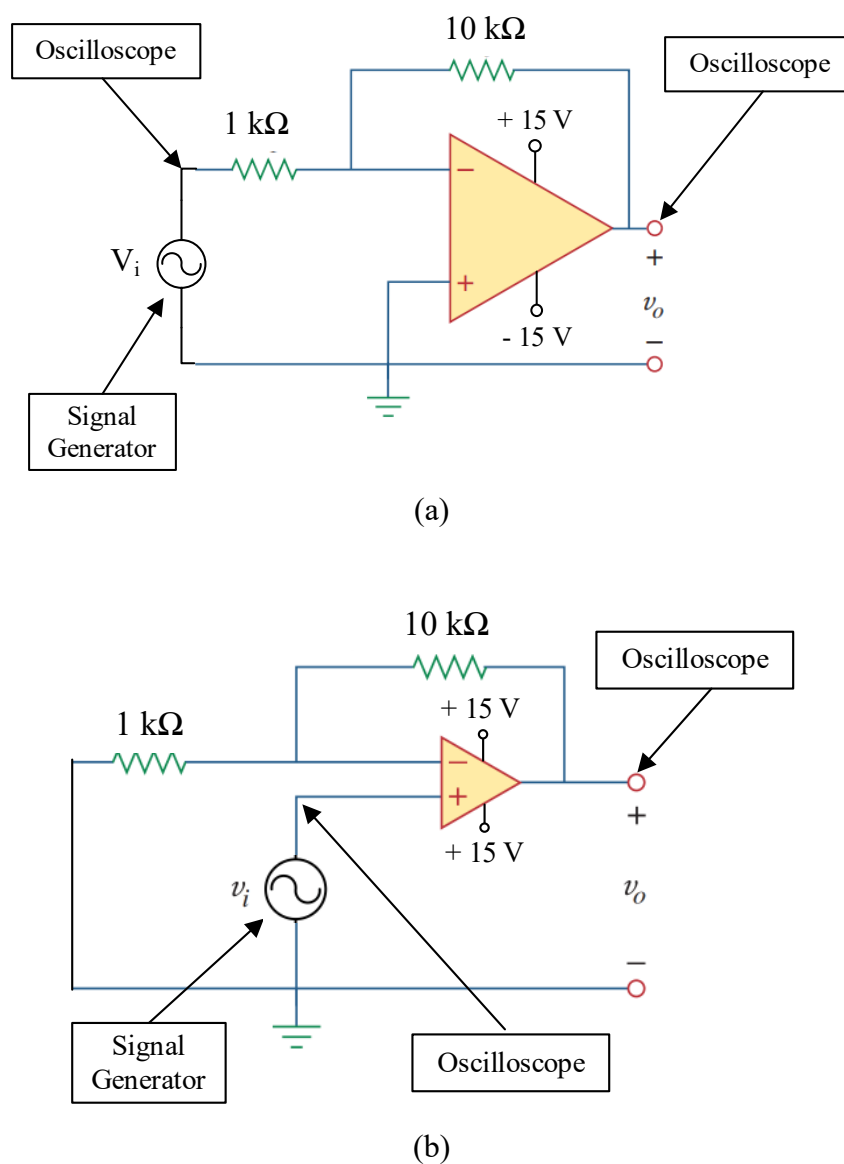


Fig. 8.3. Circuit diagram for implementing (a) inverting and (b) non-inverting amplifier circuit.

8.5. Experimental Procedure

1. Connect the circuit according to Fig. 8.3 (a).
2. Supply an ac signal from the signal generator and record the corresponding output waveform.
3. Compare the measured gain with the theoretical gain.
4. Again, connect the circuit according to Fig. 8.3 (b).
5. Repeat step-2 and 3.

8.6. Experimental Results

Attach the input and output waveforms

Table-1: Data table for comparing the experimental data with the theoretical data.

	Input Voltage, V_{in} (p-p)	Output Voltage, V_{out} (p-p)	Measured Gain ($A_v' = V_{out}/V_{in}$)	Theoretical Gain (A_v)	%Error
Inverting Amplifier					
Non-inverting Amplifier					

8.7. Discussion and Conclusion

State the concluding remarks on the experiment.

8.8. Precautions

- All the connections should be correct.
- Op-amp terminals must be identified properly.
- Calibration must be performed properly.

Experiment No.: 09

Name of the experiment: Study of differentiator and integrator circuit using Op-Amp

9.1. Objectives

- To implement the differentiator and integrator circuit using Op-Amp
- To observe the input and output waveforms of the differentiator and integrator circuit

9.2. Theory

Differentiator Circuit

A differentiator circuit produces an output voltage that is proportional to the rate of change of the input voltage. The basic Op-Amp differentiator is implemented using a capacitor in series with the input and a resistor in the feedback path.

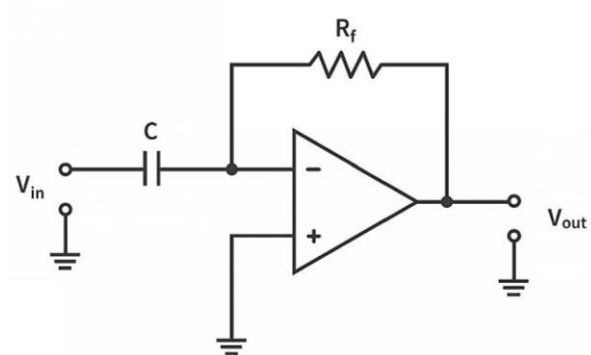


Fig. 9.1. Differentiator circuit using op-amp.

The input-output relationship is provided by:

$$V_{out} = -R_f C \frac{dV_{in}}{dt}$$

Integrator Circuit

An integrator circuit produces an output voltage that is proportional to the time integral of the input voltage. The basic Op-Amp integrator is implemented using a resistor in series with the input and a capacitor in the feedback path.

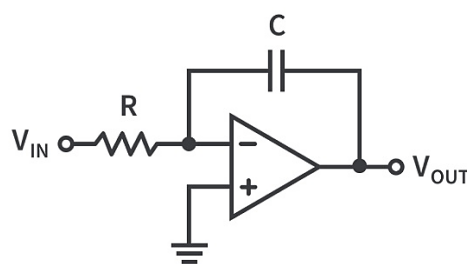


Fig. 9.2. Integrator circuit using op-amp.

The input-output relationship is provided by:

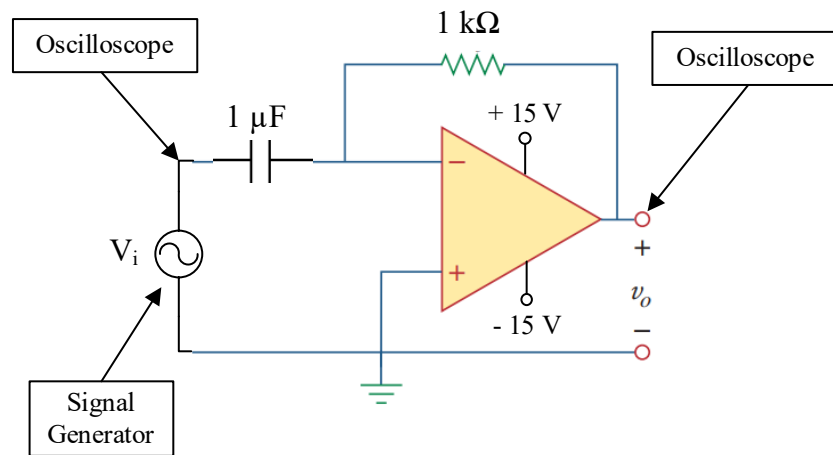
$$V_{out} = -\frac{1}{RC} \int V_{in} dt$$

9.3. Required Apparatus

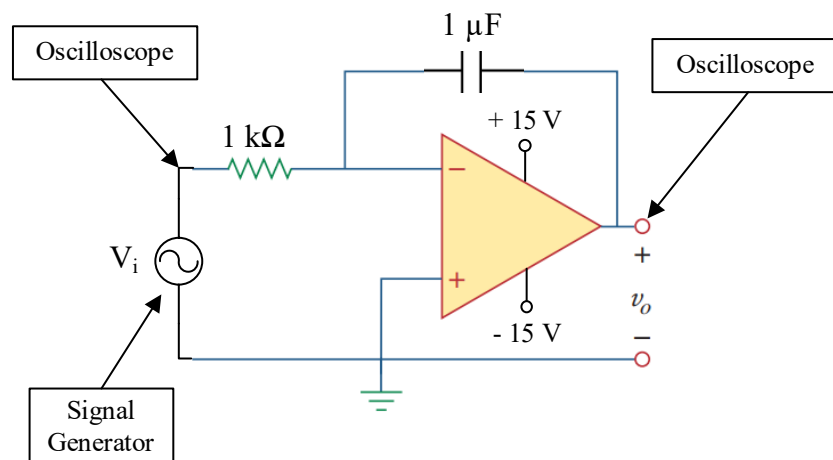
- Op-amp ($\mu c741$, 1 piece)
- Resistors (1 k Ω : 1 piece)
- Capacitor (1 μ F: 1 piece)
- Signal Generator
- Oscilloscope

- DC Power supply (0 – 15V)
- Project board
- Multimeter
- Connecting wires

9.4. Circuit Diagram



(a)



(b)

Fig. 9.3. Circuit diagram for implementing differentiator and integrator circuit.

9.5. Experimental Procedure

1. Connect the circuit according to Fig. 9.3 (a).
2. Supply an ac signal from the signal generator and record the corresponding output waveform.
3. Again, connect the circuit according to Fig. 9.3 (b).
4. Repeat step-2.

9.6. Experimental Results

Attach the input and output waveforms

9.7. Discussion and Conclusion

State the concluding remarks on the experiment.

9.8. Precautions

- All the connections should be correct.
- Op-amp terminals must be identified properly.
- Calibration must be performed properly.

Experiment No.: 10

Name of the experiment: Study of the astable multivibrator operation using 555 timer

10.1. Objectives

- To implement the astable multivibrator operation using 555 timer
- To observe the output waveforms of the astable multivibrator
- To compare the obtained frequency of oscillation with the theoretical frequency

10.2. Theory

In astable mode, the 555 timer operates as a free-running oscillator that continuously switches between high and low states, generating a square wave output. This makes it useful for applications such as clock pulses, LED flashers, and frequency generators.

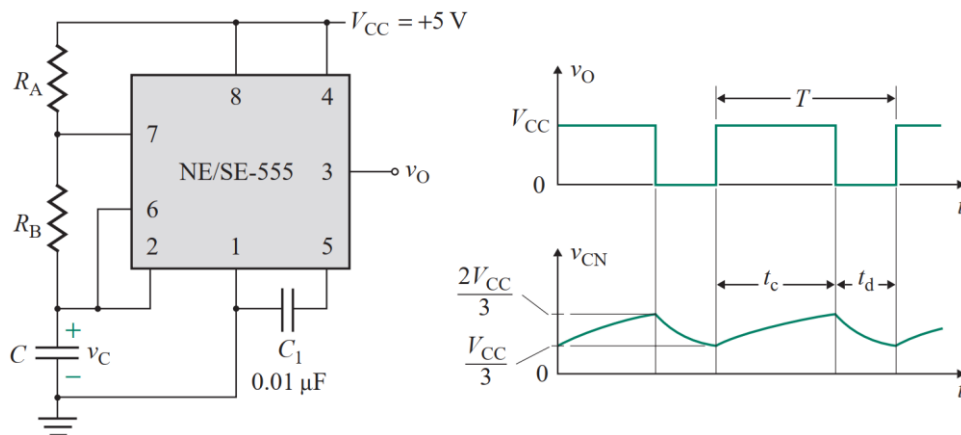


Fig. 10.1. Astable multivibrator circuit and corresponding waveforms.

The output frequency of the astable multivibrator is given by:

$$f = \frac{1.45}{(R_A + 2 R_B)C}$$

10.3. Required Apparatus

- 555-Timer (NE-555, 1 piece)
- Resistors (10 k Ω : 2 pieces)
- Capacitor (1 μF , 10 μF : 1 piece each)
- Oscilloscope
- DC Power supply (0 – 15V)
- Project board
- Multimeter
- Connecting wires

10.4. Circuit Diagram

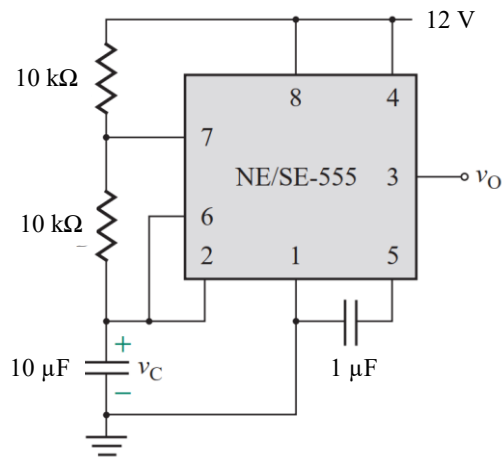
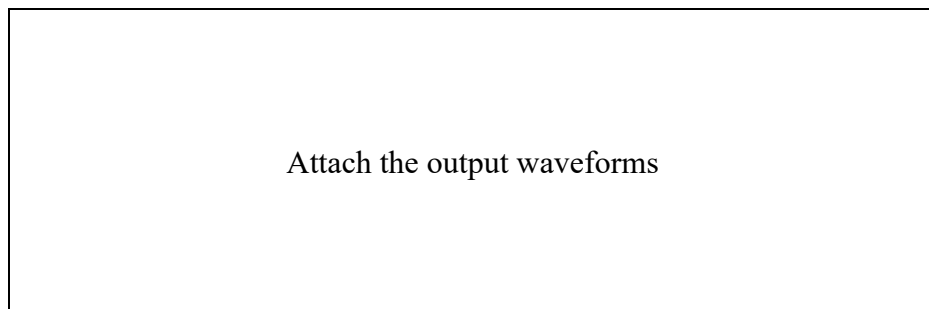


Fig. 10.2. Circuit diagram for implementing astable multivibrator circuit using 555 timer.

10.5. Experimental Procedure

1. Connect the circuit according to Fig. 10.2.
2. Supply the necessary power supply and record the corresponding output waveform.
3. Compare the obtained frequency of oscillation with the theoretical frequency.

10.6. Experimental Results



Calculation:

Theoretical frequency of oscillation,

$$f = \frac{1.45}{(R_A + 2 R_B)C}$$

Experimental frequency of oscillation,

$$f' = \frac{1}{t_{on} + t_{off}}$$

$$\%Error = \frac{|f - f'|}{f} \times 100\%$$

10.7. Discussion and Conclusion

State the concluding remarks on the experiment.

10.8. Precautions

- All the connections should be correct.
- 555 timer terminals must be identified properly.
- Calibration must be performed properly.

Experiment No.: 11

Name of the experiment: Study of constructing digital logic circuit using semiconductor devices

11.1. Objectives

- To implement a digital logic circuit using any of the semiconductor devices
- To observe the input and output waveforms of the corresponding circuit

11.2. Theory

Briefly explain the operating principle of the implemented digital logic circuit.

11.3. Required Apparatus

Enlist the required components with proper specifications to implement the circuit.

11.4. Circuit Diagram

Attach the required circuit diagram to implement the circuit

11.5. Experimental Results

Attach the input and output waveforms (if any)

Data Table:

Add a data table presenting the observed results from the experiments (if any)

11.6. Discussion and Conclusion

State the concluding remarks on the experiment.

11.7. Precautions

State the necessary precautions to be undertaken to implement the circuit